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Review

Tilapia sex determination: Where temperature and genetics meet ☆

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- It has long been known that gonad estrogens acted as natural inducers for ovarian development in lower vertebrates (Yamamoto, 1969; Nakamura et al., 1998).
- The aromatase enzyme (=Cyp19 gene) catalyzes the conversion of androgens into17β-estradiol(Baroilleretal., 1999) and if inhibited, blocks estrogen production causing a female to male sex reversal (Guiguenetal.,1999;Kwonetal.,2000).

In developing ovaries of XX fish the Cyp19a(=Cyp19a1a)geneisup-regulated.

Ijiri et al. (2008) found higher levels of Cyp19a in future ovaries as early as 9 days post fertilisation and they increased rapidly until 19.

High temperatures could efficiently masculinise some progenies if started around 10 days post fertilization (dpf) and if applied for at least 10 days, with longer periods being just as effective.

Temperature applied during the sex differentiating period in tilapia XX offspring induces a down-regulation of Cyp19a (D'Cotta et al., 2001a)seen at 17dpf.